DO NOW — RECALL

Object-oriented synoptic challenge

Name: _____ Class: _____ Date: _____

1. What is the difference between Step Into and Step Over?

2. What is encapsulation?

3. How would you store many Book objects in C#?

4. A Borrow method should fail on a book already on loan. Give a useful test.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Step Into enters the called method; Step Over executes it without stopping inside.

2. Model answer

Controlling how an object's internal state can be accessed or changed.

3. Model answer

For example List<Book>, a collection holding Book references.

4. Model answer

Set OnLoan=true, call Borrow, expect false and OnLoan to stay true.

Vocabulary — know these terms

class — Template defining properties and methods.

object — Instance of a class.

encapsulation — Control access to and modification of object state.

collection — Stores multiple elements, such as List<Book>.

invariant — A property that should stay true while an object is used.

NEW LEARNING

Read, annotate and explain

Model the entity

A Book has a Title and whether it is currently on loan.

Keep state protected

The public Borrow method updates a private-set OnLoan property, so external code cannot change it freely.

Test a second call

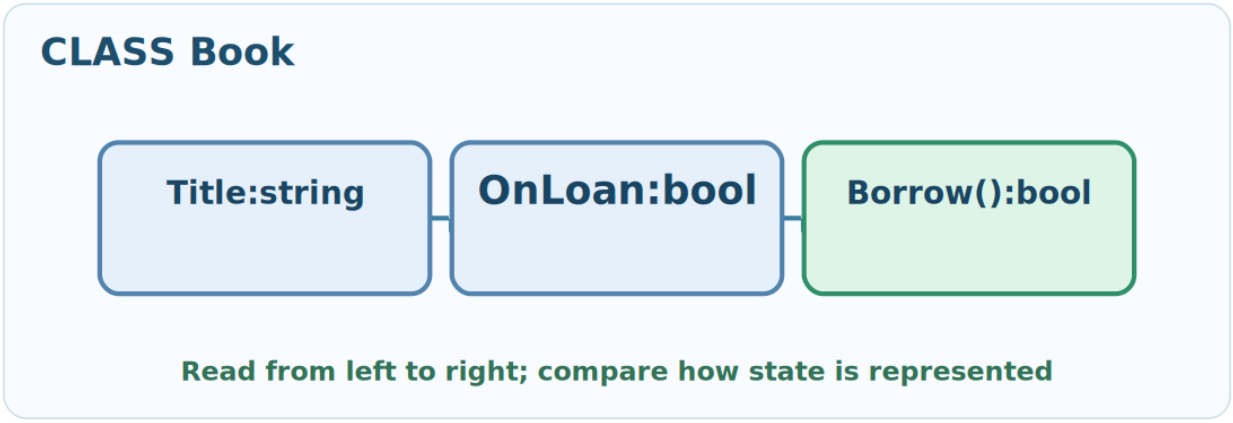
The first Borrow returns true and changes state; the second call returns false because the book is already on loan.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

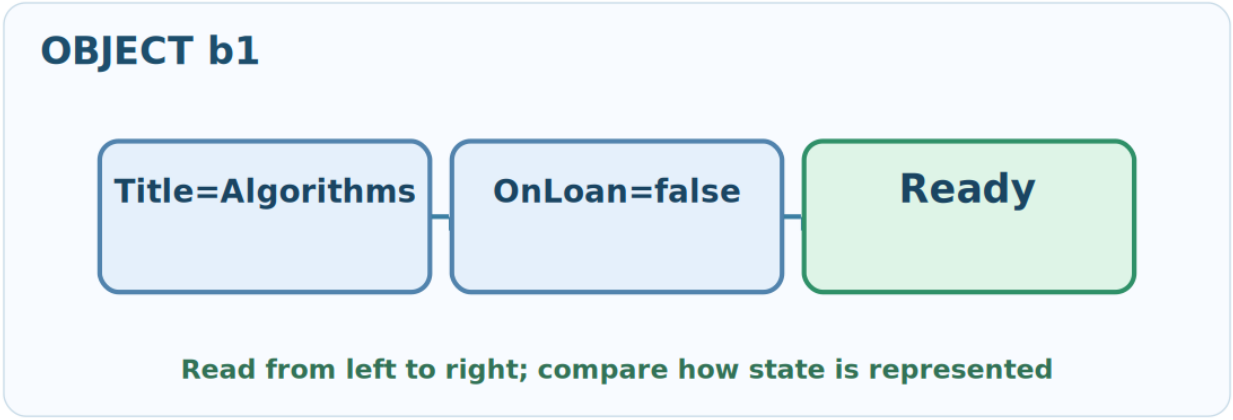
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



CLASS Book; Title:string; OnLoan:bool; Borrow():bool

New Learning

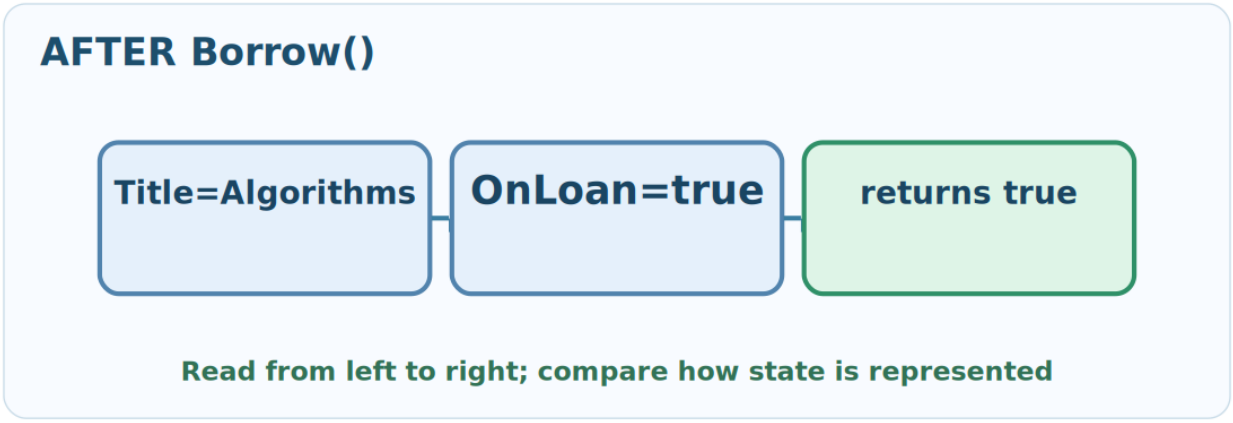


OBJECT b1; Title=Algorithms; OnLoan=false; Ready

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



AFTER Borrow(); Title=Algorithms; OnLoan=true; returns true

I DO — WORKED EXAMPLE

Follow the worked method

Model a Seat object. Reserve() should succeed once, then fail.

Predict

Initial Reserved is false, so Reserve sets it to true and returns true.

Model response

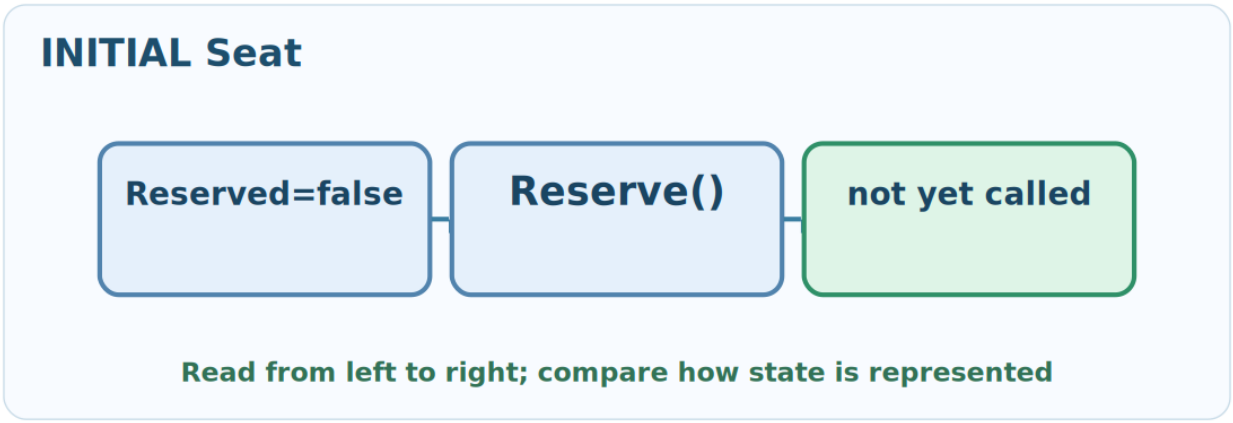
The second call detects Reserved already true; it returns false and leaves state unchanged.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Model a Seat object. Reserve() should succeed once, then fail.

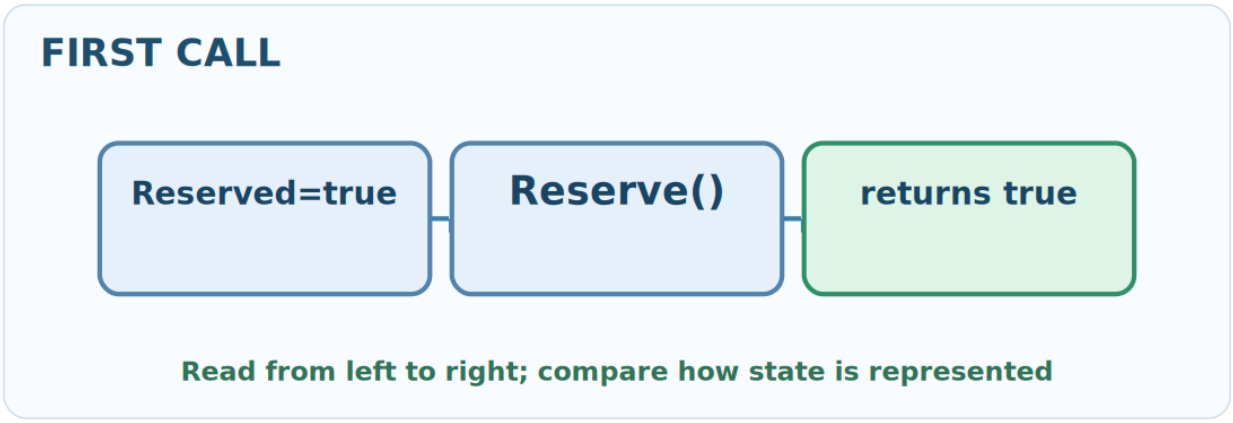
Initial Reserved is false, so Reserve sets it to true and returns true.



INITIAL Seat; Reserved=false; Reserve(); not yet called

Model a Seat object. Reserve() should succeed once, then fail.

Initial Reserved is false, so Reserve sets it to true and returns true.



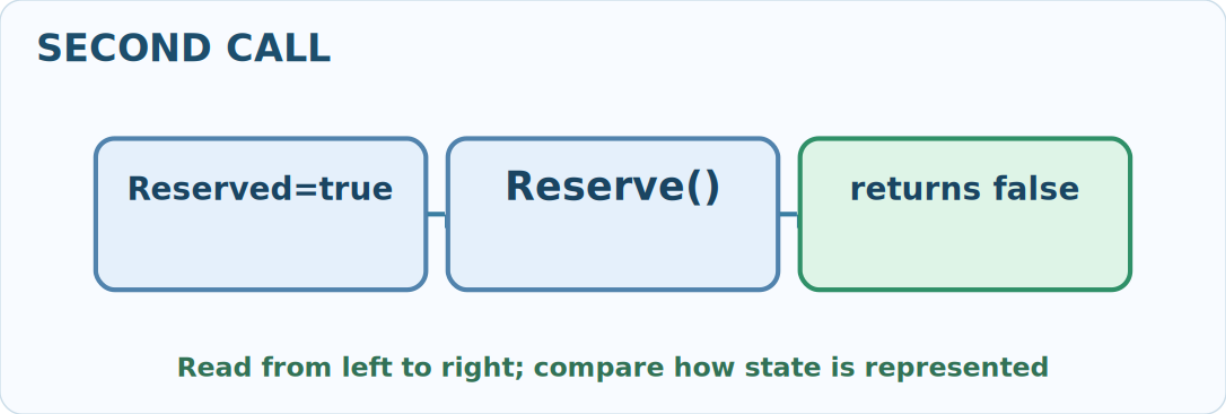
FIRST CALL; Reserved=true; Reserve(); returns true

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Model a Seat object. Reserve() should succeed once, then fail.

Initial Reserved is false, so Reserve sets it to true and returns true.



SECOND CALL; Reserved=true; Reserve(); returns false

WE DO — GUIDED PRACTICE

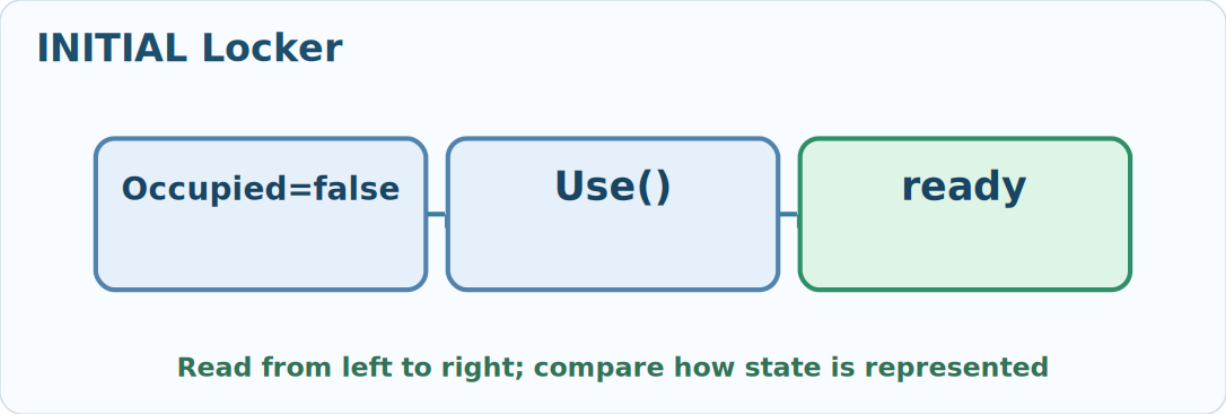
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A Locker class starts Occupied=false. What happens after Use(), then Use() again?

The first call sets Occupied to true and returns true.



INITIAL Locker; Occupied=false; Use(); ready

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict The first call sets Occupied to true and returns true.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain the difference between a class and an object using Book.

Question 2

Write a method ReturnBook() that only succeeds if OnLoan is true, then explain one test.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain how to manage many Book objects, retrieve a book by unique ISBN, and test duplicate borrowing.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Two Book objects created from the same class must always share the same OnLoan value.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

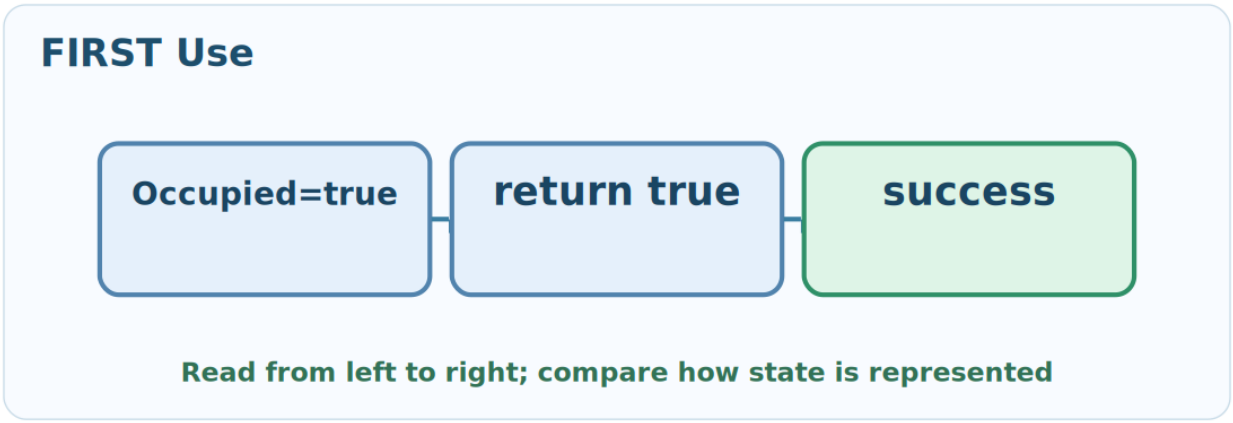
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A Locker class starts Occupied=false. What happens after Use(), then Use() again?

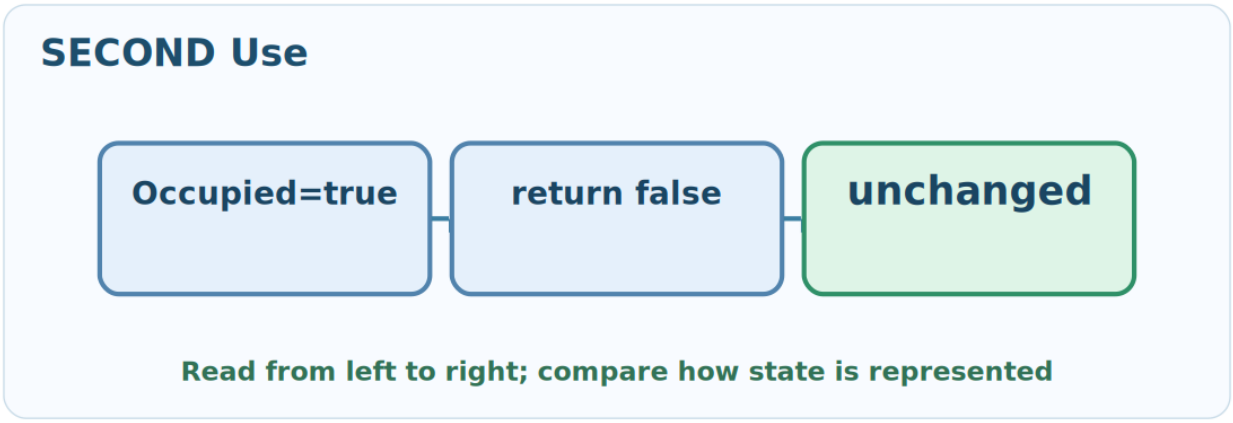
The first call sets Occupied to true and returns true.



FIRST Use; Occupied=true; return true; success

A Locker class starts Occupied=false. What happens after Use(), then Use() again?

The first call sets Occupied to true and returns true.



SECOND Use; Occupied=true; return false; unchanged

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model response

The second call returns false without changing Occupied.

IMPROVE: Explain one correction to your own reasoning.